INTERPRETABILITY FAILURE MODES on the VCS

MECHANISM BRANCH	METHOD EXHIBITING IT (phase)	F	faithfulness F vs intervention oracle (0 – 0.5 – 1.0)
1 Gradient pathologies *vanishing on the index · saturation · model-invariance* **mechanism: naive position gradient = 0 (§3.2)**	Vanilla saliency	0.38	
	SmoothGrad	0.38	
	Guided backprop	0.32	
	Grad×Input / DeepLIFT	0.49	
	Integrated Gradients	0.38	
	Expected Gradients	0.18	
2 Sampling / approximation error *surrogate variance · baseline sensitivity* **mechanism: IG baseline-sweep → branch 1**	LIME	0.64	
	KernelSHAP	0.65	
	RISE	0.55	
3 Correlational confounds *present ≠ used · spurious tuning · spurious edges* **mechanism: Hewitt & Liang control-task gap**	Tuning curves (A3)	0.21	
	Spike-word / pairwise (A4)	0.23	
	Granger causality (A6)	0.14	
	Linear probing + control (C)	0.24	
4 Decomposition non-identifiability *polysemantic / rotated atoms ≠ true variables* **mechanism: factorisation atoms not unique**	Sparse autoencoder (C)	0.15	
	NMF/PCA dictionaries (C)	0.10	
	Dim-reduction NMF (A7)	0.52	
5 Compositional blind-spots *single-site misses distributed / interacting structure* **mechanism: 1st-order / greedy-edge approx.**	Single-unit lesions (A2)	0.96	
	Attribution patching (C)	0.46	
	ACDC circuit disc. (C)	0.63	
	Path patching (C)	0.58	
0 Architecture-specific N/A (control — not a failure of method) *needs conv maps / attention / a learned policy — does not apply* **mechanism: recorded finding (§5) — nothing to score; the subject lacks the substrate**	*Grad-CAM / Grad-CAM++*	N/A	*does not apply*
	Attention rollout	N/A	*does not apply*
	VIPER	N/A	*does not apply*

(1) gradient pathologies
(2) sampling / approximation
(3) correlational confound
(4) decomposition non-id.
(5) compositional blind-spot
(0) N/A — does not apply (control)
low F (pale chip)
high F (dark chip)
95% bootstrap CI (whisker)